Functions and Relations

Binary Relations, Matrix Representations, Equivalence Relations, Functions, Bijections, and Composition

1 Binary Relations

In discrete mathematics and computer science, relationships between elements of sets—such as keys mapped to database records, vertices connected by edges in a network, or tasks constrained by scheduling dependencies—are formalized through the concept of a *relation*. While a Cartesian product $A \times B$ pairs every element of A with every element of B, a relation selects the specific subset of ordered pairs that satisfy a meaningful condition.

DEFINITION **Binary Relation**

Let A and B be sets. A **binary relation** R from set A to set B is a subset of the Cartesian product $A \times B$:

$$R \subseteq A \times B$$

- If an ordered pair (a, b) belongs to R, we write $(a, b) \in R$ or use the infix notation aRb (read as "a is related to b by R").
- If (a, b) does not belong to R, we write $(a, b) \notin R$ or $a \not R b$.
- Set A is called the **domain** of R, and set B is called the **codomain** of R.

INTUITION & DEEP DIVE **Why Relations Are Defined as Subsets of Cartesian Products**

Defining a relation R simply as a set of ordered pairs $R \subseteq A \times B$ separates *which* pairs are related from *how* the rule is phrased (such as "is a divisor of" or "is enrolled in"). Because $A \times B$ has $|A| \cdot |B|$ possible ordered pairs, there are $2^{|A| \cdot |B|}$ distinct binary relations from a finite set A to a finite set B. In relational databases (SQL), a two-column table over domains A and B is a finite binary relation $R \subseteq A \times B$.

EXAMPLE & WALKTHROUGH **Divisibility Relation Between Two Sets**

Let $P = \{2, 3, 4\}$ and $Q = \{2, 4, 8, 9, 15\}$. Suppose we define a binary relation R from P to Q by the rule:

$$(p, q) \in R \quad \text{if and only if} \quad p \text{ divides } q \quad (\text{written } p \mid q)$$

Step-by-step evaluation across $P \times Q$:

- For $p = 2$: 2 divides $2, 4, 8$ in Q, yielding pairs $(2, 2), (2, 4), (2, 8)$.
- For $p = 3$: 3 divides $9, 15$ in Q, yielding pairs $(3, 9), (3, 15)$.
- For $p = 4$: 4 divides $4, 8$ in Q, yielding pairs $(4, 4), (4, 8)$.

Collecting all valid ordered pairs gives:

$$R = \{(2, 2), (2, 4), (4, 4), (2, 8), (4, 8), (3, 9), (3, 15)\}$$

We can visualize this bipartite mapping between P and Q structurally:

Domain $P = \{2, 3, 4\}$	**Outgoing Relation Arrows to Codomain $Q = \{2, 4, 8, 9, 15\}$**
• $p = 2$ (even divisor) • $p = 3$ (odd prime divisor) • $p = 4$ (multiple of 4)	$2 \to \{2, 4, 8\}$ $3 \to \{9, 15\}$ $4 \to \{4, 8\}$

1.1 Relations on a Single Set

In many applications—such as ordering numbers, comparing strings, or modeling transitions between states of a finite automaton—the domain and codomain are the same set.

DEFINITION **Relation on a Set**

A **relation on a set A** is a binary relation from A to A, that is, a subset of $A \times A$:

$$R \subseteq A \times A$$

EXAMPLE & WALKTHROUGH **Prime Factor Relation on a Set**

Let $A = \{2, 3, 4, 8, 9\}$, and let R be the relation on A defined by:

$$(x, y) \in R \quad \text{if and only if} \quad x \text{ is a prime factor of } y$$

Analysis: For (x, y) to belong to R, x itself must be a prime number and must divide y. Among the elements of $A = \{2, 3, 4, 8, 9\}$, only 2 and 3 are prime ($4 = 2^2$, $8 = 2^3$, and $9 = 3^2$ are composite):

- For $x = 2$: 2 is a prime factor of 2, 4, and 8.
- For $x = 3$: 3 is a prime factor of 3 and 9.
- For $x \in \{4, 8, 9\}$: None of these are prime, so they contribute no pairs as first coordinates.

Therefore:

$$R = \{(2, 2), (2, 4), (2, 8), (3, 3), (3, 9)\}$$

2 Representation and Inverse of Relations

2.1 Zero-One Matrix Representation

When sets A and B are finite, a binary relation $R \subseteq A \times B$ can be represented compactly in memory as a **zero-one matrix** (boolean adjacency matrix).

CORE CONCEPT **Matrix Representation of a Binary Relation**

Suppose R is a relation from $A = \{a_1, a_2, ..., a_m\}$ to $B = \{b_1, b_2, ..., b_n\}$, where the elements of each set are listed in a fixed order. The relation R is represented by the $m \times n$ matrix $M_R = [m_{ij}]$, where:

$$M_R = \begin{pmatrix} m_{11} & m_{12} & \dots & m_{1n} \\ m_{21} & m_{22} & \dots & m_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ m_{m1} & m_{m2} & \dots & m_{mn} \end{pmatrix} \quad \text{with} \quad m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R \\ 0 & \text{if } (a_i, b_j) \notin R \end{cases}$$

Each row i corresponds to element $a_i \in A$, and each column j corresponds to element $b_j \in B$.

EXAMPLE & WALKTHROUGH **Matrix of the Divisibility Relation**

Returning to $P = \{2, 3, 4\}$ and $Q = \{2, 4, 8, 9, 15\}$ with $(p, q) \in R \iff p \mid q$, we index the elements in increasing order:

$$p_1 = 2, \quad p_2 = 3, \quad p_3 = 4$$

$$q_1 = 2, \quad q_2 = 4, \quad q_3 = 8, \quad q_4 = 9, \quad q_5 = 15$$

The relation can be written in indexed form as:

$$R = \{(2,2),(2,4),(4,4),(2,8),(4,8),(3,9),(3,15)\}$$
$$= \{(p_1,q_1),(p_1,q_2),(p_3,q_2),(p_1,q_3),(p_3,q_3),(p_2,q_4),(p_2,q_5)\}$$

Setting entry (i,j) to 1 whenever $(p_i,q_j) \in R$ and 0 otherwise yields the 3×5 zero-one matrix:

$$M = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \end{pmatrix}$$

EXAMPLE & WALKTHROUGH **Constructing a Relation Matrix**

Problem: Given $A = \{1,2,3\}$ and $B = \{a,b,c\}$, with the relation:

$$R = \{(1,a),(1,b),(2,b),(3,a),(3,b),(3,c)\}$$

Represent this relation in matrix form.

Solution: Order the domain elements as $a_1 = 1, a_2 = 2, a_3 = 3$ (rows $1,2,3$) and the codomain elements as $b_1 = a, b_2 = b, b_3 = c$ (columns $1,2,3$):

- **Row 1 (1):** $(1,a) \in R$, $(1,b) \in R$, $(1,c) \notin R \Longrightarrow [1,1,0]$.
- **Row 2 (2):** $(2,a) \notin R$, $(2,b) \in R$, $(2,c) \notin R \Longrightarrow [0,1,0]$.
- **Row 3 (3):** $(3,a) \in R$, $(3,b) \in R$, $(3,c) \in R \Longrightarrow [1,1,1]$.

Combining the three rows gives the 3×3 matrix representation:

$$M_R = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

2.2 Inverse Relations

Reversing the direction of every ordered pair in a relation R from A to B produces a relation from B to A.

DEFINITION **Inverse Relation**

Let R be a binary relation from set A to set B. The **inverse relation** of R, denoted by R^{-1}, is the relation from B to A defined by reversing the order of every pair in R:

$$R^{-1} = \{(b,a) \mid (a,b) \in R\}$$

In matrix representation, the matrix of R^{-1} is the **transpose** of the matrix of R:

$$M_{R^{-1}} = (M_R)^T$$

EXAMPLE & WALKTHROUGH **Inverse of the Divisibility Relation**

Let $P = \{2,3,4\}$ and $Q = \{2,4,8,9,15\}$, where $(p,q) \in R$ if p divides q:

$$R = \{(2,2),(2,4),(4,4),(2,8),(4,8),(3,9),(3,15)\}$$

The inverse relation $R^{-1} \subseteq Q \times P$ is characterized verbally by:

$$(q,p) \in R^{-1} \quad \text{if and only if} \quad q \text{ is a multiple of } p$$

Reversing each ordered pair in R yields:

$$R^{-1} = \{(2,2),(4,2),(4,4),(8,2),(8,4),(9,3),(15,3)\}$$

Its matrix representation is the 5×3 transpose M^T:

$$M_{R^{-1}} = M^T = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

3 Properties of Relations on a Set

When a relation R is defined on a single set A ($R \subseteq A \times A$), we classify its behavior using four fundamental properties: *reflexivity*, *symmetry*, *antisymmetry*, and *transitivity*.

3.1 Reflexivity

DEFINITION **Reflexive Relation**

A relation R on a set A is called **reflexive** if every element of A is related to itself:

$$\forall a \in A, \quad (a, a) \in R$$

Matrix signature: Every entry on the main diagonal of M_R is 1 (i.e., $m_{ii} = 1$ for all $1 \leq i \leq |A|$).

EXAMPLE & WALKTHROUGH **Testing Reflexivity on a Finite Set**

Let $A = \{1, 2, 3\}$ and consider the relation:

$$R = \{(1,1), (1,3), (2,2), (2,1), (2,3), (3,3)\}$$

Because A contains the elements $1, 2, 3$ and R contains all three self-loops $(1,1) \in R$, $(2,2) \in R$, and $(3,3) \in R$, the relation R is **reflexive**.

EXAMPLE & WALKTHROUGH **Testing Whether y = x² Is Reflexive on Positive Integers**

Problem: A relation R is defined on the set of positive integers $\mathbb{Z}^+$ by:

$$R : y = x^2, \quad x, y \in \mathbb{Z}^+$$

Is this relation reflexive?

Solution: No, R is not reflexive. By definition, for R to be reflexive on $\mathbb{Z}^+$, the pair (x, x) must belong to R for *every* positive integer $x \in \mathbb{Z}^+$. That requires $x = x^2$ for all $x \in \mathbb{Z}^+$.

- For $x = 1$: $1 = 1^2 = 1$, so $(1,1) \in R$.
- For $x = 2$: $2^2 = 4 \neq 2$, so $(2,2) \notin R$ (instead $(2,4) \in R$).

Since the universal requirement $\forall x \in \mathbb{Z}^+, (x, x) \in R$ fails for $x = 2$, R is *not reflexive*.

3.2 Symmetry and Antisymmetry

DEFINITION **Symmetric Relation**

A relation R on a set A is called **symmetric** if whenever a is related to b, b is also related to a:

$$\forall a, b \in A, \quad (a, b) \in R \implies (b, a) \in R$$

Matrix signature: M_R is a symmetric matrix ($M_R = (M_R)^T$, meaning $m_{ij} = m_{ji}$ for all i, j).

EXAMPLE & WALKTHROUGH **Symmetric Relation**

Let $A = \{1, 2, 3, 4\}$ and let R be defined as:

$$R = \{(1, 1), (1, 2), (2, 1), (2, 3), (3, 2), (4, 4)\}$$

Every off-diagonal pair has its reverse present in R: $(1, 2)$ is paired with $(2, 1)$, and $(2, 3)$ is paired with $(3, 2)$. Therefore, R is **symmetric**. Note that R is *not* reflexive because $(2, 2) \notin R$ and $(3, 3) \notin R$.

DEFINITION **Antisymmetric Relation**

A relation R on a set A is called **antisymmetric** if two distinct elements are never mutually related—that is, if both $(a, b) \in R$ and $(b, a) \in R$ hold, then a and b must be the same element:

$$\forall a, b \in A, \quad ((a, b) \in R \land (b, a) \in R) \implies a = b$$

Equivalently, by contraposition: for all $a \neq b$, if $(a, b) \in R$, then $(b, a) \notin R$. **Matrix signature:** For all $i \neq j$, at most one of m_{ij} and m_{ji} can be 1 (i.e., $m_{ij} \cdot m_{ji} = 0$ for $i \neq j$). Diagonal entries m_{ii} may be 0 or 1 freely.

EXAMPLE & WALKTHROUGH **Antisymmetric Relation**

Let $A = \{1, 2, 3\}$ and let $R = \{(1, 1), (2, 2), (1, 2)\}$.

- For the self-loops $(1, 1)$ and $(2, 2)$, $a = b$ holds automatically.
- For the distinct pair $(1, 2) \in R$, the reverse pair $(2, 1)$ is *not* in R.

Hence there is no pair of distinct elements $a \neq b$ with both $(a, b) \in R$ and $(b, a) \in R$, which means R is **antisymmetric**.

PITFALL / WARNING **Symmetric and Antisymmetric Are Not Opposites**

It is easy to assume that "not symmetric" means "antisymmetric," or that a relation cannot be both. In fact:

1. **A relation can be both symmetric and antisymmetric:** Any diagonal relation (such as the equality relation $R = \{(1, 1), (2, 2), (3, 3)\}$ or the empty relation $R = \varnothing$) is both symmetric and antisymmetric.
2. **A relation can be neither symmetric nor antisymmetric:** For example, $R = \{(1, 2), (2, 1), (2, 3)\}$ is not symmetric (missing $(3, 2)$) and not antisymmetric (contains both $(1, 2)$ and $(2, 1)$).

3.3 Transitivity

DEFINITION **Transitive Relation**

A relation R on a set A is called **transitive** if whenever a is related to b and b is related to c, then a is also directly related to c:

$$\forall a, b, c \in A, \quad ((a, b) \in R \land (b, c) \in R) \implies (a, c) \in R$$

EXAMPLE & WALKTHROUGH **Verifying Transitivity via Two-Step Chains**

Let $A = \{1, 2, 3, 4\}$ and define $R = \{(1, 2), (1, 3), (2, 3), (3, 4), (1, 4), (2, 4)\}$. To verify that R is transitive, we inspect every composable pair $(a, b) \in R$ and $(b, c) \in R$ and confirm that (a, c) belongs to R:

First Pair (a, b)	Second Pair (b, c)	Required (a, c)	Status in R
$(1, 2)$	$(2, 3)$	$(1, 3)$	Present in R (✓)
$(1, 2)$	$(2, 4)$	$(1, 4)$	Present in R (✓)
$(1, 3)$	$(3, 4)$	$(1, 4)$	Present in R (✓)
$(2, 3)$	$(3, 4)$	$(2, 4)$	Present in R (✓)

Because every two-step path $a \to b \to c$ has its direct edge $a \to c$ in R (in fact, R is the strict ordering $<$ on $\{1, 2, 3, 4\}$), R is **transitive**.

EXAMPLE & WALKTHROUGH **Analyzing the Relation x + y = 5 on {1, 2, 3, 4}**

Problem: Let $A = \{1, 2, 3, 4\}$ and let relation R on A be defined by:

$$R : x + y = 5 \quad \text{for } x, y \in A$$

Determine all properties of relation R.

Solution: First, list all ordered pairs $(x, y) \in A \times A$ satisfying $x + y = 5$:

$$R = \{(1, 4), (2, 3), (3, 2), (4, 1)\}$$

Now test each of the four properties:

1. *Reflexive?* **No.** For $x = 1$, $1 + 1 = 2 \neq 5$, so $(1, 1) \notin R$ (no (a, a) is in R because $2a = 5$ has no integer solution).
2. *Symmetric?* **Yes.** Addition is commutative ($x + y = 5 \implies y + x = 5$). Explicitly, $(1, 4), (4, 1) \in R$ and $(2, 3), (3, 2) \in R$.
3. *Antisymmetric?* **No.** Both $(1, 4) \in R$ and $(4, 1) \in R$ hold with $1 \neq 4$.
4. *Transitive?* **No.** We have $(1, 4) \in R$ and $(4, 1) \in R$, which by transitivity would require $(1, 1) \in R$, but $(1, 1) \notin R$.

Conclusion: The relation R is **symmetric only** (it is neither reflexive, nor antisymmetric, nor transitive).

3.4 Equivalence Relations

DEFINITION **Equivalence Relation**

A relation R on a set A is called an **equivalence relation** if it simultaneously satisfies three properties:

1. **Reflexive**: $(a, a) \in R$ for all $a \in A$.
2. **Symmetric**: $(a, b) \in R \implies (b, a) \in R$ for all $a, b \in A$.
3. **Transitive**: $(a, b) \in R \land (b, c) \in R \implies (a, c) \in R$ for all $a, b, c \in A$.

INTUITION & DEEP DIVE **Equivalence Relations Partition a Set into Disjoint Classes**

An equivalence relation generalizes "equality" by grouping elements that share a chosen attribute (such as congruence modulo n, having the same string length, or residing in the same connected component of an undirected graph). Every equivalence relation R on A partitions A into mutually disjoint subsets called *equivalence classes* $[a]_R = \{x \in A \mid (a, x) \in R\}$. Within each class, every element is related to every other element.

EXAMPLE & WALKTHROUGH **Equivalence Relation on A = {1, 2, 3, 4}**

Let $A = \{1, 2, 3, 4\}$ and consider the relation:

$$R = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3), (4,4)\}$$

Let us check why R is an equivalence relation:

- **Reflexive:** $(1,1), (2,2), (3,3), (4,4) \in R$.
- **Symmetric:** Every pair among $\{1,2,3\}$ has its reverse in R, and $(4,4)$ is its own reverse.
- **Transitive:** Notice that $R = (\{1,2,3\} \times \{1,2,3\}) \cup \{(4,4)\}$. Any composable pairs (a,b) and (b,c) either both lie inside $\{1,2,3\} \times \{1,2,3\}$ (which contains all 9 pairs on $\{1,2,3\}$, so $(a,c) \in R$ automatically) or both equal $(4,4)$.

Thus R is an **equivalence relation** that partitions A into two equivalence classes: $[1] = [2] = [3] = \{1,2,3\}$ and $[4] = \{4\}$.

Summary of the Four Core Relation Properties on a Set A

Reflexive

Condition: $\forall a \in A, (a,a) \in R$.
Graph: Every vertex has a self-loop.
Matrix: All diagonal entries $m_{ii} = 1$.

Symmetric

Condition: $(a,b) \in R \implies (b,a) \in R$.
Graph: Every directed edge between distinct vertices is bidirectional.
Matrix: $M_R = (M_R)^T$.

Antisymmetric

Condition: $(a,b) \in R \wedge (b,a) \in R \implies a = b$.
Graph: No two *distinct* vertices have two-way arrows between them.
Matrix: If $i \neq j$, then $m_{ij} = 0$ or $m_{ji} = 0$.

Transitive

Condition: $(a,b) \in R \wedge (b,c) \in R \implies (a,c) \in R$.
Graph: Every 2-step path $a \to b \to c$ has a direct shortcut edge $a \to c$.
Combine: Reflexive + Symmetric + Transitive = **Equivalence Relation**.

4 Functions: Fundamentals and Specifications

A *function* is a binary relation in which every input from the domain is assigned to a single, unambiguous output in the codomain.

DEFINITION **Function / Mapping / Transformation**

Let A and B be non-empty sets. A binary relation f from A to B is a **function** (also called a **mapping** or **transformation**) if every element in A is associated with **exactly one** element in B. We write:

$$f : A \to B$$

which reads "f maps A to B." Formally, $f \subseteq A \times B$ is a function if and only if:

1. **Totality (Every input is mapped):** $\forall a \in A, \exists b \in B$ such that $(a,b) \in f$.
2. **Uniqueness (At most one output per input):** $\forall a \in A, \forall b_1, b_2 \in B$, if $(a,b_1) \in f$ and $(a,b_2) \in f$, then $b_1 = b_2$.

CORE CONCEPT **Domain, Codomain, Image, Pre-image, and Range**

When $f : A \to B$ and $(a, b) \in f$, we write $f(a) = b$:

- **Domain:** The set A of all valid inputs.
- **Codomain:** The target set B of potential outputs.
- **Image:** If $f(a) = b$, then b is the *image* of a under f.
- **Pre-image:** If $f(a) = b$, then a is a *pre-image* of b under f.
- **Range:** The set of all actual images attained by f, denoted $f(A)$ or $\text{Range}(f)$:

$$\text{Range}(f) = f(A) = \{f(a) \mid a \in A\} = \{b \in B \mid \exists a \in A, f(a) = b\} \subseteq B$$

The range of f is always a subset (possibly a *proper subset*) of the codomain B.

4.1 Specifying a Function

A function can be specified explicitly in two main ways:

1. **Set of Ordered Pairs:** For finite sets $A = \{1, 2, 3\}$ and $B = \{u, v, w\}$, we can list the pairs directly:

$$f = \{(1, u), (2, v), (3, w)\} \iff f(1) = u, \quad f(2) = v, \quad f(3) = w$$

2. **Assignment Formula:** We can provide an algebraic rule or expression for computing $f(x)$ from x, such as:

$$f(x) = x^2, \quad \text{or equivalently in set-builder form:} \quad f = \{(x, x^2) \mid x \in \mathbb{R}\}$$

EXAMPLE & WALKTHROUGH **Determining Whether a Relation Is a Function**

Problem: Given $A = \{1, 2, 3\}$, $B = \{u, v, w\}$, and $f = \{(1, u), (2, u), (2, v), (3, w)\}$. Is f a function?

Solution: No, f is not a function. Let us check each element of the domain $A = \{1, 2, 3\}$:

- Element $1 \in A$ is mapped only to u.
- Element $2 \in A$ appears in *two distinct ordered pairs*: $(2, u) \in f$ and $(2, v) \in f$, where $u \neq v$.
- Element $3 \in A$ is mapped only to w.

Because the input $2 \in A$ is associated with two different outputs in B (u and v), f violates the uniqueness condition. Note that having $(1, u)$ and $(2, u)$ share the output u is completely fine for a function, but mapping a single input 2 to two outputs u and v is not allowed.

5 Classification of Functions: Injection, Surjection, Bijection

Functions $f : A \to B$ are classified by whether distinct inputs stay distinct in the output and whether the range covers the entire codomain B.

5.1 Injection (One-to-One Functions)

DEFINITION **Injection / One-to-One Function**

A function $f : A \to B$ is called **one-to-one** or **injective** (an **injection**) if distinct elements of A always map to distinct elements of B:

$$\forall a, b \in A, \quad a \neq b \implies f(a) \neq f(b)$$

By contraposition, this is equivalent to:

$$\forall a, b \in A, \quad f(a) = f(b) \implies a = b$$

In an arrow diagram, every element of B receives *at most one* incoming arrow from A. For finite sets, an injection $f : A \to B$ can exist only if $|A| \leq |B|$.

EXAMPLE & WALKTHROUGH **Testing Injectivity of f(x) = x²**

Problem:

1. Is the function $f : \mathbb{Z} \to \mathbb{Z}$ defined by $f(x) = x^2$ injective?
2. What if we define $f = \{(x, x^2) \mid x \geq 0\}$ (restricting the domain to non-negative integers)?

Solution:

1. *For $f : \mathbb{Z} \to \mathbb{Z}$ with $f(x) = x^2$:* **Not injective.** Take $a = 1 \in \mathbb{Z}$ and $b = -1 \in \mathbb{Z}$. Then $a \neq b$, yet:

$$f(1) = 1^2 = 1 \quad \text{and} \quad f(-1) = (-1)^2 = 1 \implies f(1) = f(-1)$$

   Because two distinct integers share the same image, f is not one-to-one on $\mathbb{Z}$.
2. *When restricted to $x \geq 0$ (i.e., $f : \mathbb{N}_0 \to \mathbb{Z}$ where $f = \{(x, x^2) \mid x \geq 0\}$):* **Yes, f is injective.** Suppose $a, b \geq 0$ and $f(a) = f(b)$. Then:

$$a^2 = b^2 \implies a^2 - b^2 = 0 \implies (a - b)(a + b) = 0$$

   Since $a \geq 0$ and $b \geq 0$, $a + b = 0$ can only happen when $a = b = 0$; in all other cases $a + b > 0$, forcing $a - b = 0$, so $a = b$. Restricting the domain removes the negative duplicates and makes f one-to-one.

5.2 Surjection (Onto Functions)

DEFINITION **Surjection / Onto Function**

A function $f : A \to B$ is called **onto** or **surjective** (a **surjection**) if every element of the codomain B is the image of *at least one* element of the domain A:

$$\forall b \in B, \quad \exists a \in A \quad \text{such that} \quad f(a) = b$$

Equivalently, the range of f equals the entire codomain: $\text{Range}(f) = f(A) = B$. In an arrow diagram, every element of B receives *one or more* incoming arrows from A. For finite sets, a surjection $f : A \to B$ can exist only if $|A| \geq |B|$.

EXAMPLE & WALKTHROUGH **Testing Surjectivity of f(x) = x⁴ + 3**

Problem: Is the function $f : \mathbb{Z} \to \mathbb{Z}$ defined by $f(x) = x^4 + 3$ onto?

Solution: No, f is not onto. For $f : \mathbb{Z} \to \mathbb{Z}$ to be onto, for every integer $y \in \mathbb{Z}$ there would need to exist an integer $x \in \mathbb{Z}$ such that $x^4 + 3 = y$, or $x^4 = y - 3$.

- *Lower bound on the range:* Since $x^4 \geq 0$ for all $x \in \mathbb{Z}$, we have $f(x) = x^4 + 3 \geq 3$ for all $x \in \mathbb{Z}$. Thus, no integer $y < 3$ (for example, $y = 0$, which would require $x^4 = -3$) is in the range of f.
- *Gaps above 3:* Even for $y \geq 3$, take $y = 5 \in \mathbb{Z}$. Then $x^4 + 3 = 5 \implies x^4 = 2$, which has no integer solution $x \in \mathbb{Z}$.

Because $\text{Range}(f) = \{3, 4, 19, 84, \ldots\} \subsetneq \mathbb{Z}$, f is not onto.

5.3 Bijection and Inverse Functions

DEFINITION **Bijection / One-to-One Correspondence**

A function $f : A \to B$ is a **bijection** (or **one-to-one correspondence**) if it is simultaneously:

1. **Injective (one-to-one)**: $f(a) = f(b) \implies a = b$, and
2. **Surjective (onto)**: $\forall b \in B, \exists a \in A$ with $f(a) = b$.

Every element in B is the image of a *unique* element in A. For finite sets, a bijection between A and B exists if and only if $|A| = |B|$.

DEFINITION **Inverse Function**

Let $f : A \to B$ be a **bijection**. The **inverse function** of f, denoted by $f^{-1} : B \to A$, is the function that assigns to each element $b \in B$ the unique element $a \in A$ such that $f(a) = b$:

$$f^{-1}(b) = a \iff f(a) = b$$

A function $f : A \to B$ is *invertible* (has an inverse function $f^{-1} : B \to A$) if and only if f is a bijection.

PITFALL / WARNING **Inverse Relation R^{-1} vs. Inverse Function f^{-1}**

Any binary relation $R \subseteq A \times B$ always has an *inverse relation* $R^{-1} \subseteq B \times A$. However, if $f : A \to B$ is a function, its inverse relation $f^{-1} = \{(b, a) \mid (a, b) \in f\}$ is a valid *function* from B to A **if and only if f is a bijection**:

- If f is *not injective*, then some $b \in B$ has two pre-images $a_1 \neq a_2$, so $f^{-1}(b)$ would have two outputs (violating uniqueness).
- If f is *not surjective*, then some $b \in B$ has *no* pre-image in A, so $f^{-1}(b)$ would be undefined on b (violating totality).

EXAMPLE & WALKTHROUGH **Computing Inverse Functions**

1. **Finite Set of Ordered Pairs:** Let $A = \{1, 2, 3\}$, $B = \{u, v, w\}$, and $f = \{(1, u), (2, v), (3, w)\}$. Since f is both one-to-one and onto, its inverse function is obtained by swapping each pair:
$$f^{-1} = \{(u, 1), (v, 2), (w, 3)\}$$
2. **Algebraic Formula:** Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x + 1$. Since f is a bijection, we find f^{-1} by solving $y = f(x)$ for x:
$$y = x + 1 \implies x = y - 1 \implies f^{-1}(y) = y - 1 \quad (\text{or } f^{-1}(x) = x - 1)$$

6 Function Composition

When the codomain of one function matches (or is contained within) the domain of a second function, we can chain the two mappings together into a pipeline.

DEFINITION **Composition of Functions**

Let $g : A \to B$ be a function from set A to set B, and let $f : B \to C$ be a function from set B to set C. The **composition** of f and g, denoted by $f \circ g$, is the function from A to C defined by:

$$(f \circ g)(a) = f(g(a)) \quad \text{for all } a \in A$$



Function Composition Pipeline: g is applied first (A → B), followed by f (B → C)

EXAMPLE & WALKTHROUGH **Composing Two and Three Functions**

Problem: Given $f(x) = x + 1$ and $g(x) = x^2$ on $\mathbb{R}$:

1. Compute $(f \circ g)(x)$ and compare it with $(g \circ f)(x)$.
2. Given $h(x) = x + 1$, determine the triple composition $(f \circ g \circ h)(x)$.

Solution:

1. **Two-function compositions:** Applying g first and then f:

$$(f \circ g)(x) = f(g(x)) = f(x^2) = x^2 + 1$$

   By contrast, applying f first and then g:

$$(g \circ f)(x) = g(f(x)) = g(x + 1) = (x + 1)^2 = x^2 + 2x + 1$$

   Notice that $(f \circ g)(x) \neq (g \circ f)(x)$—function composition is **not commutative** in general.

2. **Triple composition $(f \circ g \circ h)(x)$ with $h(x) = x + 1$:** Function composition is *associative*, and we evaluate from right to left (innermost function first):

$$\begin{aligned}(f \circ g \circ h)(x) &= f(g(h(x))) \\ &= f(g(x + 1)) \quad (\text{since } h(x) = x + 1) \\ &= f((x + 1)^2) \quad (\text{since } g(u) = u^2) \\ &= (x + 1)^2 + 1 \quad (\text{since } f(v) = v + 1) \\ &= x^2 + 2x + 2\end{aligned}$$

7 Worked Practice Problems

Below are complete step-by-step solutions to practice problems on injectivity, surjectivity, and bijections.

EXAMPLE & WALKTHROUGH **Injectivity and Surjectivity on $\mathbb{Z} \to \mathbb{Z}$**

Problem:

- **Part 1:** Determine whether each of these functions from $\mathbb{Z}$ to $\mathbb{Z}$ is one-to-one (injective): **(a)** $f(n) = n - 1$ **(b)** $f(n) = n^2 + 1$ **(c)** $f(n) = n^3$ **(d)** $f(n) = \lceil \frac{n}{2} \rceil$
- **Part 2:** Which of the functions above are onto (surjective)?

Solutions:

- **(a) $f(n) = n - 1$:**
  - *One-to-one?* **Yes.** Suppose $f(a) = f(b)$ for $a, b \in \mathbb{Z}$. Then $a - 1 = b - 1 \implies a = b$.
  - *Onto?* **Yes.** Let $y \in \mathbb{Z}$ be any target integer. Choose $n = y + 1 \in \mathbb{Z}$. Then $f(y + 1) = (y + 1) - 1 = y$. Since f is both one-to-one and onto, $f(n) = n - 1$ is a *bijection* on $\mathbb{Z}$.
- **(b) $f(n) = n^2 + 1$:**
  - *One-to-one?* **No.** Counterexample: $f(1) = 1^2 + 1 = 2$ and $f(-1) = (-1)^2 + 1 = 2$, so $f(1) = f(-1)$ with $1 \neq -1$.
  - *Onto?* **No.** Since $n^2 \geq 0$ for all $n \in \mathbb{Z}$, $f(n) = n^2 + 1 \geq 1$. Thus no integer $y \leq 0$ (e.g., $y = 0$) is in the range of f (nor is $y = 3$, since $n^2 = 2$ has no integer solution).
- **(c) $f(n) = n^3$:**
  - *One-to-one?* **Yes.** The cube function is strictly increasing on $\mathbb{R}$ (and hence on $\mathbb{Z}$): if $a^3 = b^3$, taking real cube roots yields $a = b$.

- *Onto?* **No.** Only perfect cubes belong to the range. For example, $y = 2 \in \mathbb{Z}$ has no integer pre-image because $n^3 = 2$ has no solution in $\mathbb{Z}$ ($1^3 = 1 < 2 < 8 = 2^3$).

- **(d) $f(n) = \lceil \frac{n}{2} \rceil$:**
  - *One-to-one?* **No.** Every consecutive pair of odd and even integers maps to the same ceiling value: for example, $f(1) = \lceil \frac{1}{2} \rceil = 1$ and $f(2) = \lceil \frac{2}{2} \rceil = 1$, yet $1 \neq 2$.
  - *Onto?* **Yes.** Given any target integer $y \in \mathbb{Z}$, choose the even integer $n = 2y \in \mathbb{Z}$ (or $n = 2y - 1$). Then $f(2y) = \lceil \frac{2y}{2} \rceil = \lceil y \rceil = y$.

Function $f : \mathbb{Z} \to \mathbb{Z}$	One-to-One	Onto	Bijection?
(a) $f(n) = n - 1$	Yes (✓)	Yes (✓)	Yes
(b) $f(n) = n^2 + 1$	No (⊗, $f(1) = f(-1)$)	No (⊗, $f(n) \geq 1$)	No
(c) $f(n) = n^3$	Yes (✓)	No (⊗, $2 \notin$ Range)	No
(d) $f(n) = \lceil \frac{n}{2} \rceil$	No (⊗, $f(1) = f(2)$)	Yes (✓, $f(2y) = y$)	No

EXAMPLE & WALKTHROUGH **Surjectivity of Two-Variable Integer Functions f : ℤ × ℤ → ℤ (Set I)**

Problem: Determine whether $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is onto if: **(a)** $f(m, n) = 2m - n$ **(b)** $f(m, n) = m^2 - n^2$ **(c)** $f(m, n) = m + n + 1$ **(d)** $f(m, n) = |m| - |n|$ **(e)** $f(m, n) = m^2 - 4$

Solutions: To show $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is *onto*, we check whether every integer $k \in \mathbb{Z}$ can be written as $f(m, n) = k$ for some $(m, n) \in \mathbb{Z} \times \mathbb{Z}$.

- **(a) $f(m, n) = 2m - n$: Onto.** For any $k \in \mathbb{Z}$, set $m = 0$ and $n = -k$ (both in $\mathbb{Z}$). Then $f(0, -k) = 2(0) - (-k) = k$.

- **(b) $f(m, n) = m^2 - n^2$: Not onto.** Factoring gives $m^2 - n^2 = (m - n)(m + n)$. Notice that $(m - n)$ and $(m + n)$ always have the *same parity* (their difference is $2n$, which is even).
  - If both factors are even, their product is a multiple of 4.
  - If both factors are odd, their product is odd.

  Therefore, $m^2 - n^2$ can never equal an integer congruent to 2 modulo 4. Specifically, $k = 2 \in \mathbb{Z}$ has no integer solution because the only factorizations of 2 are 1×2 and $(-1) \times (-2)$, where the factors have opposite parity. Thus $2 \notin \text{Range}(f)$.

- **(c) $f(m, n) = m + n + 1$: Onto.** For any $k \in \mathbb{Z}$, choose $m = k - 1 \in \mathbb{Z}$ and $n = 0 \in \mathbb{Z}$. Then $f(k - 1, 0) = (k - 1) + 0 + 1 = k$.

- **(d) $f(m, n) = |m| - |n|$: Onto.** Let $k \in \mathbb{Z}$ be any integer:
  - If $k \geq 0$, choose $(m, n) = (k, 0)$, giving $f(k, 0) = |k| - |0| = k$.
  - If $k < 0$, choose $(m, n) = (0, k)$, giving $f(0, k) = |0| - |k| = -(-k) = k$.

  Every integer $k \in \mathbb{Z}$ is achieved.

- **(e) $f(m, n) = m^2 - 4$: Not onto.** Since $m^2 \geq 0$ for all $m \in \mathbb{Z}$, $f(m, n) = m^2 - 4 \geq -4$. No integer $k < -4$ (such as $k = -5$) is in the range (nor is $k = -2$, since $m^2 = 2$ has no integer solution).

EXAMPLE & WALKTHROUGH **Surjectivity of Two-Variable Integer Functions f : ℤ × ℤ → ℤ (Set II)**

Problem: Determine whether the function $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is onto if: **(a)** $f(m, n) = m + n$ **(b)** $f(m, n) = m^2 + n^2$ **(c)** $f(m, n) = m$ **(d)** $f(m, n) = |n|$ **(e)** $f(m, n) = m - n$

Solutions:

- **(a) $f(m, n) = m + n$: Onto.** For any $k \in \mathbb{Z}$, choose $(m, n) = (k, 0) \in \mathbb{Z} \times \mathbb{Z}$; then $f(k, 0) = k + 0 = k$.
- **(b) $f(m, n) = m^2 + n^2$: Not onto.** Since $m^2 + n^2 \geq 0$, negative integers (e.g., $k = -1$) are never attained, and positive integers like $k = 3$ cannot be written as a sum of two integer squares.
- **(c) $f(m, n) = m$: Onto.** This is the projection onto the first coordinate. For any $k \in \mathbb{Z}$, $f(k, 0) = k$.
- **(d) $f(m, n) = |n|$: Not onto.** Since $|n| \geq 0$ for all $n \in \mathbb{Z}$, no negative integer (such as $k = -1$) is in the range of f.
- **(e) $f(m, n) = m - n$: Onto.** For any $k \in \mathbb{Z}$, choose $(m, n) = (k, 0) \in \mathbb{Z} \times \mathbb{Z}$; then $f(k, 0) = k - 0 = k$.

EXAMPLE & WALKTHROUGH **Testing Bijections from ℝ to ℝ**

Problem: Determine whether each of these functions is a bijection from $\mathbb{R}$ to $\mathbb{R}$:

- **Set 1: (a)** $f(x) = -3x + 4$ **(b)** $f(x) = -3x^2 + 7$ **(c)** $f(x) = \frac{x+1}{x+2}$ **(d)** $f(x) = x^5 + 1$
- **Set 2: (a)** $f(x) = 2x + 1$ **(b)** $f(x) = x^2 + 1$ **(c)** $f(x) = x^3$ **(d)** $f(x) = \frac{x^2+1}{x^2+2}$

Solutions for Set 1:

- **1(a) $f(x) = -3x + 4$: Bijection.**
  - *One-to-one:* $-3a + 4 = -3b + 4 \implies -3a = -3b \implies a = b$.
  - *Onto:* For any $y \in \mathbb{R}$, solving $y = -3x + 4$ gives $x = \frac{4-y}{3} \in \mathbb{R}$, so $f\left(\frac{4-y}{3}\right) = y$.
- **1(b) $f(x) = -3x^2 + 7$: Not a bijection** (neither one-to-one nor onto).
  - *Not one-to-one:* $f(1) = 4 = f(-1)$ with $1 \neq -1$.
  - *Not onto:* Since $x^2 \geq 0$, $f(x) \leq 7$, so $y = 8 \in \mathbb{R}$ has no real pre-image.
- **1(c) $f(x) = \frac{x+1}{x+2}$: Not a bijection from $\mathbb{R}$ to $\mathbb{R}$.**
  - *Not defined on all of* $\mathbb{R}$*:* At $x = -2$, the denominator is 0, so $f(-2)$ is undefined.
  - *Not onto* $\mathbb{R}$*:* Rewriting $f(x) = \frac{(x+2)-1}{x+2} = 1 - \frac{1}{x+2}$ shows that $f(x) \neq 1$ for all $x \in \mathbb{R} \setminus \{-2\}$, so $y = 1$ has no pre-image. (Note: f is a bijection from $\mathbb{R} \setminus \{-2\}$ to $\mathbb{R} \setminus \{1\}$.)
- **1(d) $f(x) = x^5 + 1$: Bijection.**
  - *One-to-one:* $a^5 + 1 = b^5 + 1 \implies a^5 = b^5 \implies a = b$ (since odd powers are strictly increasing on $\mathbb{R}$).
  - *Onto:* For any $y \in \mathbb{R}$, every real number has a unique real fifth root $x = \sqrt[5]{y - 1} \in \mathbb{R}$, satisfying $f(\sqrt[5]{y - 1}) = (y - 1) + 1 = y$.

Solutions for Set 2:

- **2(a) $f(x) = 2x + 1$: Bijection.**
  - *One-to-one:* $2a + 1 = 2b + 1 \implies a = b$.
  - *Onto:* For any $y \in \mathbb{R}$, $x = \frac{y-1}{2} \in \mathbb{R}$ satisfies $f\left(\frac{y-1}{2}\right) = y$ (with inverse $f^{-1}(y) = \frac{y-1}{2}$).
- **2(b) $f(x) = x^2 + 1$: Not a bijection** (neither one-to-one nor onto).
  - *Not one-to-one:* $f(1) = 2 = f(-1)$.
  - *Not onto:* $f(x) = x^2 + 1 \geq 1$, so $y = 0 \in \mathbb{R}$ has no real pre-image.
- **2(c) $f(x) = x^3$: Bijection.**
  - *One-to-one:* $a^3 = b^3 \implies a = b$ in $\mathbb{R}$.
  - *Onto:* For any $y \in \mathbb{R}$, $x = \sqrt[3]{y} \in \mathbb{R}$ satisfies $f(\sqrt[3]{y}) = (\sqrt[3]{y})^3 = y$. (Unlike on $\mathbb{Z}$, $f(x) = x^3$ *is* onto on $\mathbb{R}$.)
- **2(d) $f(x) = \frac{x^2+1}{x^2+2}$: Not a bijection** (neither one-to-one nor onto).

- *Not one-to-one:* $f(1) = \frac{2}{3} = f(-1)$ since $(-x)^2 = x^2$.
- *Not onto:* Writing $f(x) = 1 - \frac{1}{x^2+2}$ and using $x^2 \geq 0 \implies x^2 + 2 \geq 2 \implies 0 < \frac{1}{x^2+2} \leq \frac{1}{2}$, we see that $\text{Range}(f) = \left[\frac{1}{2}, 1\right) \subsetneq \mathbb{R}$.